

Problem-2 Schmidt decomposition

a)

$$a) |\psi\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Solution

$$|\psi\rangle = \frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

$$= \frac{1}{\sqrt{2}} |0\rangle|0\rangle + \frac{1}{\sqrt{2}} |1\rangle|1\rangle$$

Schmidt Coefficient: $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ ✓

Schmidt Rank = 2 $\Rightarrow$ Entangled

Schmidt vectors

$$\begin{aligned} |u_1\rangle &= |0\rangle, & |u_2\rangle &= |1\rangle \\ |v_1\rangle &= |0\rangle, & |v_2\rangle &= |1\rangle \end{aligned} \quad \left. \vphantom{\begin{aligned} |u_1\rangle &= |0\rangle, \\ |v_1\rangle &= |0\rangle, \end{aligned}} \right\}$$

$$b) |\psi\rangle = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}$$

Solution

$$|\psi\rangle = \frac{1}{2} (|0\rangle(|0\rangle + |1\rangle)) + \frac{1}{2} (|1\rangle(|0\rangle + |1\rangle))$$

$$= \frac{1}{2} (|0\rangle + |1\rangle)(|0\rangle + |1\rangle)$$

$$= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle \otimes |+\rangle$$

$$\underline{\underline{|\psi\rangle = |+\rangle \otimes |+\rangle}}$$

Schmidt Coefficient: 1

Schmidt Rank: 1 $\Rightarrow$ Separable

Schmidt vectors

$$|u\rangle = |+\rangle, \quad |v\rangle = |+\rangle, \quad \perp$$

c) Solution

$$C = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}$$

$$C^T C = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix} \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} \frac{2}{3} & \frac{2}{3} \\ \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

Calculate eigenvalue

$$\det(C^T C - \lambda I) = 0$$

$$\det \begin{pmatrix} \frac{2}{3} - \lambda & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} - \lambda \end{pmatrix} = 0$$

$$\det \begin{pmatrix} \frac{2-3\lambda}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2-3\lambda}{3} \end{pmatrix} = 0$$

$$\frac{(2-3\lambda)(2-3\lambda) - 1}{9} = 0$$

$$(2-3\lambda)(2-3\lambda) - 1 = 0$$

$$2 - 6\lambda - 3\lambda + 9\lambda^2 - 1 = 0$$

$$1 - 9\lambda + 9\lambda^2 = 0$$

$$\lambda^2 - \lambda + \frac{1}{9} = 0$$

Quadratic form

$$\lambda = \frac{2 \pm \sqrt{2^2 - 4 \times \frac{1}{9} \times 1}}{2 \times 1}$$

$$\lambda = \frac{2 \pm \sqrt{2 - \frac{4}{9}}}{2}$$

$$\lambda = \frac{2 \pm \sqrt{\frac{9-4}{9}}}{2}$$

$$= \frac{2 \pm \sqrt{\frac{5}{9}}}{2}$$

$$= \frac{2 \pm \frac{1}{3}\sqrt{5}}{2}$$

$$= \frac{3 \pm \sqrt{5}}{2 \times 3}$$

$$= \frac{3 \pm \sqrt{5}}{6}$$

$$\lambda_1 = \frac{3+\sqrt{5}}{2} \quad \left. \begin{array}{l} \lambda_1 = \frac{3+\sqrt{5}}{2} \\ \lambda_2 = \frac{3-\sqrt{5}}{2} \end{array} \right\} \text{The two eigen values.}$$

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Eigenvector for λ_1

$$(C - \lambda_1 I) \vec{v}_0 = 0$$

$$\frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \lambda_1 \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 3\lambda_1 \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

Therefore

$$2v_1 + v_2 = 3\lambda_1 v_1$$

$$v_1 + v_2 = 3\lambda_1 v_2$$

$$v_2 = 3\lambda_1 v_1 - 2v_1$$

$$= \left(3 \times \left(\frac{3+\sqrt{5}}{2} \right) - 2 \right) v_1$$

$$= \left(\frac{3+\sqrt{5}}{2} - 2 \right) v_1$$

$$= \left(\frac{3+\sqrt{5}-4}{2} \right) v_1$$

$$= \left(\frac{-1+\sqrt{5}}{2} \right) v_1$$

Normalized $\vec{v}_0$

$$\vec{v}_0 = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{-1+\sqrt{5}}{2} \end{pmatrix}$$

$$\text{norm: } \|\vec{v}_0\| = \sqrt{1^2 + \left(\frac{-1+\sqrt{5}}{2} \right)^2} = \sqrt{\frac{2+3-\sqrt{5}}{2}}$$

$$= \sqrt{\frac{5-\sqrt{5}}{2}}$$

$\vec{v}_0$ normalized:

$$\frac{1}{\sqrt{5-\sqrt{5}}} \begin{pmatrix} 1 \\ \frac{-1+\sqrt{5}}{2} \end{pmatrix} = \frac{\sqrt{2}}{\sqrt{5-\sqrt{5}}} \begin{pmatrix} 1 \\ \frac{-1+\sqrt{5}}{2} \end{pmatrix}$$

Eigen vector for A_2

$$\frac{1}{3} \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \lambda_0 \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

Normalized

$$\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = 3\lambda_1 \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

$$\vec{u}_1 = \begin{pmatrix} 2 \\ \frac{-2 - \sqrt{3}}{2} \end{pmatrix}$$

$$V_2 = 3\lambda_1 u_1 - 2u_1$$

$$= \left(3 \times \left(\frac{3 - \sqrt{3}}{6} \right) - 2 \right) u_1$$

$$= \left(\frac{3 - \sqrt{3}}{2} - 2 \right) u_1$$

$$= \left(\frac{-2 - \sqrt{3}}{2} \right) u_1$$

Norm:

$$\|\vec{u}_1\| = \sqrt{\frac{5 + \sqrt{3}}{2}}$$

Normalized u_1 :

$$u_1 = \sqrt{\frac{2}{5 + \sqrt{3}}} \begin{pmatrix} 2 \\ \frac{-2 - \sqrt{3}}{2} \end{pmatrix}$$

The Schmidt decomposition

$$|\psi\rangle_{AB} = \beta_0 |u_0\rangle_A \otimes |u_0\rangle_B + \beta_1 |u_1\rangle_A \otimes |u_1\rangle_B$$

$$= \sqrt{\frac{3 + \sqrt{3}}{6}} |u_0\rangle_A \otimes |u_0\rangle_B + \sqrt{\frac{3 - \sqrt{3}}{6}} |u_1\rangle_A \otimes |u_1\rangle_B$$

As the same

$$|\psi\rangle_{AB} = \sum_{i,j} \lambda_{ij} |i\rangle_A \otimes |j\rangle_B$$

5)

$$= \frac{2 + \sqrt{5}}{\sqrt{2}(2 + 2\sqrt{5})} \left(2|0\rangle + (2 - \sqrt{5})|2\rangle \right) \otimes \left(2|0\rangle + (2 - \sqrt{5})|2\rangle \right)$$

$$|| 1$$

2) If the state is separable the Schmidt rank must be 1

$|\psi\rangle \text{ is separable } \Leftrightarrow \text{Schmidt rank of } |\psi\rangle = 1$

3) Reduced density matrix of subsystem A

ⓐ ρ_A is mixed $\Leftrightarrow |\psi\rangle$ is Entangled

$$\rho_A = \text{Tr}_B(|\psi\rangle\langle\psi|)$$

$$= \text{Tr}_B \left(\sum_{i,j} \lambda_i \lambda_j^* |\psi_i\rangle\langle\psi_j|_A \otimes |\psi_i\rangle\langle\psi_j|_B \right)$$

$$= \sum_{i,j} \lambda_i \lambda_j^* |\psi_i\rangle\langle\psi_j|_A \text{Tr}_B(|\psi_i\rangle\langle\psi_j|_B)$$

$$= \sum_{i,j} \lambda_i \lambda_j^* |\psi_i\rangle\langle\psi_j|_A \left(\langle\psi_i|_B |\psi_j\rangle_B \right) = \sum_{i,j} \lambda_i \lambda_j^* |\psi_i\rangle\langle\psi_j|_A$$

$$= \sum_{i=j} |\lambda_i|^2 |\psi_i\rangle\langle\psi_i|$$

If $|\psi\rangle$ is Entangled

$$\rho_A = \sum_{i=1}^d |\lambda_i|^2 |\psi_i\rangle\langle\psi_i|$$

Calculated the purities

$$\text{Tr}(\rho_A) = \sum_i |\lambda_i|^2 \cdot |\lambda_i|^2 = \sum_{i=1}^d |\lambda_i|^4$$

$$\text{Tr}(\rho_A^2) = \sum_{i=1}^d |\lambda_i|^4$$

If $\lambda_i \neq 0$, rank = 2, $\text{Tr}(\rho_A^2) < (\text{Tr}(\rho_A))^2$ (pure)

$\lambda_i \neq 0$, $\sum_i |\lambda_i|^2 < 1 \Rightarrow \rho_A$ mixed, ρ_A mixed $\Leftrightarrow |\psi\rangle$ is Entangled

2) Given

$$|\psi\rangle_{A_1} = \alpha|0\rangle + \beta|1\rangle$$

$$|\psi\rangle_{A_2B} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)_{A_2B}$$

Solution

$$|\psi\rangle = |\psi\rangle_{A_1} \otimes |\psi\rangle_{A_2B}$$

$$= (\alpha|0\rangle + \beta|1\rangle)_{A_1} \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)_{A_2B}$$

$$= \frac{1}{\sqrt{2}} [\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle]_{A_1A_2B}$$

$$\rightarrow |00\rangle_{A_1A_2} = \frac{1}{\sqrt{2}} [|\phi^+\rangle + |\phi^-\rangle] \quad |01\rangle_{A_1A_2} = \frac{1}{\sqrt{2}} [|\psi^+\rangle + |\psi^-\rangle]$$

$$\rightarrow |10\rangle_{A_1A_2} = \frac{1}{\sqrt{2}} [|\psi^+\rangle - |\psi^-\rangle] \quad |11\rangle_{A_1A_2} = \frac{1}{\sqrt{2}} [|\phi^+\rangle + |\phi^-\rangle]$$

Write $|\psi\rangle$ in Bell state for bases A_1A_2

$$|\psi\rangle = \frac{1}{2} [\alpha [|\phi^+\rangle + |\phi^-\rangle]_{A_1A_2} |0\rangle_B + \beta [|\psi^+\rangle - |\psi^-\rangle]_{A_1A_2} |0\rangle_B + \alpha [|\psi^+\rangle + |\psi^-\rangle]_{A_1A_2} |1\rangle_B + \beta [|\phi^+\rangle - |\phi^-\rangle]_{A_1A_2} |1\rangle_B]$$

$$= \frac{1}{2} [|\phi^+\rangle_{A_1A_2} [\alpha|0\rangle + \beta|1\rangle]_B + |\phi^-\rangle_{A_1A_2} [\alpha|0\rangle - \beta|1\rangle]_B + |\psi^+\rangle_{A_1A_2} [\alpha|1\rangle + \beta|0\rangle]_B + |\psi^-\rangle_{A_1A_2} [\alpha|1\rangle - \beta|0\rangle]_B]$$

~~$$|\psi\rangle = \frac{1}{2} [|\phi^+\rangle_{A_1A_2} (\alpha|0\rangle + \beta|1\rangle)_B + |\phi^-\rangle_{A_1A_2} (\alpha|0\rangle - \beta|1\rangle)_B + |\psi^+\rangle_{A_1A_2} (\alpha|1\rangle + \beta|0\rangle)_B + |\psi^-\rangle_{A_1A_2} (\alpha|1\rangle - \beta|0\rangle)_B]$$~~

$$|\psi\rangle = \frac{1}{2} [|\phi^+\rangle_{A_1A_2} (\alpha|0\rangle + \beta|1\rangle)_B + |\phi^-\rangle_{A_1A_2} (\alpha|0\rangle - \beta|1\rangle)_B + |\psi^+\rangle_{A_1A_2} (\alpha|1\rangle + \beta|0\rangle)_B + |\psi^-\rangle_{A_1A_2} (\alpha|1\rangle - \beta|0\rangle)_B]$$

$$\underline{\underline{|\psi^-\rangle_{A_1A_2} (\alpha|1\rangle - \beta|0\rangle)_B}}$$

2. What is the smallest number of classical bits one needs to send Bob?

Solution

Alice must send 2 classical bits to Bob by classical communication, because Her BSM has 4 possible outcomes and $\log_2 4 = 2$ bits are needed to uniquely specify which outcome occurred.

3. Once Alice communicates her measurement outcome to Bob, Bob applies corrective operation on his qubit to recover the original state $|\psi\rangle_B = \alpha|0\rangle_B + \beta|1\rangle_B$

Solution

From problem we get the Alice BSM

$$|\psi\rangle = \frac{1}{2} [|\phi^+\rangle_{A_1 A_2} (\alpha|0\rangle + \beta|1\rangle)_B + |\phi^-\rangle_{A_1 A_2} (\alpha|0\rangle - \beta|1\rangle)_B + |\psi^+\rangle_{A_1 A_2} (\alpha|1\rangle + \beta|0\rangle)_B + |\psi^-\rangle_{A_1 A_2} (\alpha|1\rangle - \beta|0\rangle)_B]$$

Conditional State for Bob depending on Alice measurement

1. Alice measure: $|\phi^+\rangle_{A_1 A_2}$

Bob's qubit: $\alpha|0\rangle + \beta|1\rangle$

Correction: I (do nothing)

2. Alice measure: $|\phi^-\rangle_{A_1 A_2}$

Bob's qubit state: $|\psi\rangle = \alpha|0\rangle - \beta|1\rangle$

Correction: Z (phase flip)

$$Z(|\psi\rangle) = Z(\alpha|0\rangle - \beta|1\rangle)$$

$$= \underline{\underline{\alpha|0\rangle + \beta|1\rangle}}$$

3. Alice measure: $|\psi^+\rangle_{A_1 A_2}$

Bob's qubit: $|\psi\rangle = \beta|0\rangle + \alpha|1\rangle$

Correction: α (bit flip)

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$$\begin{aligned} X|\psi\rangle &= X(\beta|0\rangle + \alpha|1\rangle) \\ &= \beta(X|0\rangle) + \alpha(X|1\rangle) \\ &= \beta|1\rangle + \alpha|0\rangle \\ &= \alpha|0\rangle + \beta|1\rangle \\ &= \underline{\underline{\alpha|0\rangle + \beta|1\rangle}} \end{aligned}$$

4) Alice measures $|\psi\rangle_{AA'}$

Bob's input: $|\psi\rangle = -\beta|0\rangle + \alpha|1\rangle$

Correction: X^2 (Both bit and phase flip)

$$\begin{aligned} X^2|\psi\rangle &= X(-\beta|0\rangle + \alpha|1\rangle) \\ &= X(-\beta|0\rangle - \alpha|1\rangle) \\ &= -(X\beta|0\rangle + X\alpha|1\rangle) \\ &= -(\beta|1\rangle + \alpha|0\rangle) \\ &= \underline{\underline{-(\alpha|0\rangle + \beta|1\rangle)}} \end{aligned}$$

NB: Since the negative is global phase, which does not affect the physical measurement.

Summary

$$|\psi\rangle = \frac{1}{2} [|\phi^+\rangle_{AA'} (\underbrace{(\alpha|0\rangle + \beta|1\rangle)}_{=I|\phi\rangle_B})_B + |\phi^-\rangle_{AA'} (\underbrace{(\alpha|0\rangle - \beta|1\rangle)}_{=Z|\phi\rangle_B})_B]$$

$$+ |\psi^+\rangle_{AA'} (\underbrace{(\alpha|1\rangle + \beta|0\rangle)}_{=X|\phi\rangle_B})_B + |\psi^-\rangle_{AA'} (\underbrace{(\alpha|1\rangle - \beta|0\rangle)}_{=XZ|\phi\rangle_B})_B]$$

$$|\psi\rangle = \frac{1}{2} [|\phi^+\rangle_{AA'} |\phi\rangle_B + |\phi^-\rangle_{AA'} Z|\phi\rangle_B + |\psi^+\rangle_{AA'} X|\phi\rangle_B + |\psi^-\rangle_{AA'} XZ|\phi\rangle_B]$$

After Bob with Pauli correction/operation the teleportation procedure is complete.

4) Given
 U -gate

Required
 $U|\psi\rangle$ - Show this final state of Bob?

Solution

It is very nice procedure/protocol to give soln.

Go step by step by follow the protocols

1) - Alice holds the qubit $(|\psi\rangle)$ - representation

$$|\psi\rangle_{A_1} = \alpha|0\rangle + \beta|1\rangle$$

2) Alice and Bob prepare the ebit $(I_A \otimes U_B)|\Phi^+\rangle_{AB}$
 $(|\Phi^+\rangle)$

$$|\psi\rangle_{A_2B} = (I_A \otimes U_B)|\Phi^+\rangle_{AB}$$

3). Alice measures her two qubits in the Bell state

$$|\psi\rangle = |\psi\rangle_{A_1} \otimes |\psi\rangle_{A_2B}$$

$$= (\alpha|0\rangle + \beta|1\rangle)_{A_1} \otimes (I_A \otimes U_B) \left(\frac{|00\rangle + |11\rangle}{\sqrt{2}} \right)_{A_2B}$$

$$= \frac{1}{\sqrt{2}} [\alpha|00\rangle + \alpha|01\rangle + \alpha|10\rangle \otimes U_B|0\rangle + \beta|10\rangle \otimes U_B|1\rangle + \beta|11\rangle \otimes U_B|0\rangle + \beta|11\rangle \otimes U_B|1\rangle]_{A_1 A_2 B}$$

$$= \frac{1}{2} [\alpha(|\Phi^+\rangle + |\Phi^-\rangle)_{A_1 A_2} U_B|0\rangle_B +$$

9)

1) Solution

2 - Alice qubit: $|\psi\rangle$

2 - Shared entangled qubit: $(I \otimes U)|\phi\rangle$

3 - Alice measure her two qubits in the Bell basis

lets - put down clearly for easy calculation.

Alice qubit:

$$|\psi\rangle_{A1}$$

Shared qubit:

$$|\phi\rangle_{A2B}$$

$$|\psi\rangle_{A1} \otimes |\phi\rangle_{A2B} = |\psi\rangle_{A2} \otimes (I_{A2} \otimes U_B) |\phi\rangle_{A2B}$$

$$= |\psi\rangle_{A2} \otimes (I_{A2} \otimes U_B) \left[\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right]_{A2B}$$

$$= \frac{1}{\sqrt{2}} \left(|\psi\rangle_{A2} \otimes |0\rangle_{A2} \otimes U_B |0\rangle_B + |\psi\rangle_{A2} \otimes |1\rangle_{A2} \otimes U_B |1\rangle_B \right)$$

Expressed in the Bell basis

$$|\psi\rangle_{A2} \otimes (I_{A2} \otimes U_B) |\phi\rangle_{A2B} = \frac{1}{2} \left[|00\rangle_{A2A2} \otimes U_B |\psi\rangle_B + |01\rangle_{A2A2} \otimes U_B |\psi\rangle_B + |10\rangle_{A2A2} \otimes U_B |\psi\rangle_B + |11\rangle_{A2A2} \otimes U_B |\psi\rangle_B \right]$$

$$= \frac{1}{2} \sum_i |\Phi_i\rangle_{A2A2} \otimes U_B \sigma_i |\psi\rangle_B$$

4) Measurement and correction

Alice sends $|\psi\rangle_{A2} \otimes (I_{A2} \otimes U_B) |\phi\rangle_{A2B}$ and Bob apply measurement

~~Bob's~~ σ_i

$$H_{22} = (I_{2 \times 2} \otimes U_B \sigma_i U_B^\dagger) |\psi\rangle_{A2} \otimes (I_{A2} \otimes U_B) |\phi\rangle_{A2B}$$

$$= (I_{2 \times 2} \otimes U_B \sigma_i U_B^\dagger) \frac{1}{2} \sum_i |\Phi_i\rangle_{A2A2} \otimes U_B \sigma_i |\psi\rangle_B$$

$$= \frac{1}{2} \sum_i |\Phi_i\rangle_{A2A2} \otimes U_B \sigma_i U_B^\dagger U_B \sigma_i |\psi\rangle_B$$

$$|\psi_2\rangle = \frac{1}{2} \sum_{i=0}^3 |\phi_i\rangle_{A_1 A_2} \otimes U_B \delta_i^0 \underbrace{U_B^\dagger U_B}_{=I} \delta_i^1 |\psi\rangle_B$$

$$= \frac{1}{2} \sum_{i=0}^3 |\phi_i\rangle_{A_1 A_2} \otimes U_B \delta_i^0 I \delta_i^1 |\psi\rangle_B$$

where A_1

$$\delta_i^2 = I$$

$$|\psi_2\rangle = \frac{1}{2} \sum_{i=0}^3 |\phi_i\rangle_{A_1 A_2} \otimes U_B \delta_i^2 \underbrace{\delta_i^1}_{=I} |\psi\rangle_B$$

$$= \frac{1}{2} \sum_{i=0}^3 |\phi_i\rangle_{A_1 A_2} \otimes U_B |\psi\rangle_B$$

$$= \frac{1}{2} \sum_{i=0}^3 |\phi_i\rangle_{A_1 A_2} \otimes U |\psi\rangle_B$$

Therefore The final state of Bob measurement is $U|\psi\rangle_B$ & Proof

5- Teleportation is completed

$$(U \delta_i^0 U^\dagger)_B (|\psi\rangle_{A_1} \otimes (I_{A_2} \otimes U)) |\phi_i\rangle_{AB} = U |\psi\rangle$$

Problem-3: Superdense coding

(2)

Solution

$$|\phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)_{AB}$$

Alice Acts a local unitary operation on her qubits with one of the four Pauli operation on the $|\phi^+\rangle$ to the four Bell state.

What Alice do??

- ⊗ Alice and Bob has share Entangled Bell state $|\phi^+\rangle_{AB}$ in there lab.
- ⊗ Alice want send 2 classical bit information by one quantum state
- ⊗ To send the classical bit information Alice must apply local operation (Pauli operation only on her qubit) from one of the four bases on her qubits.

Pauli transformation

General equation: $(\sigma \otimes I) |\phi^+\rangle_{AB}$

$$1. (\hat{\sigma}_I \otimes I) |\phi^+\rangle = |\phi^+\rangle$$

$$2. (\hat{\sigma}_x \otimes I) |\phi^+\rangle = |\psi^+\rangle$$

$$3. (\hat{\sigma}_z \otimes I) |\phi^+\rangle = |\phi^-\rangle$$

$$4. \left\{ \begin{array}{l} (\hat{\sigma}_y \otimes I) |\phi^+\rangle = |\psi^-\rangle \\ (-i\hat{\sigma}_y \otimes I) |\phi^+\rangle = |\psi^-\rangle \end{array} \right\}$$

Alice's unitary operation to encoding the information apply Pauli operation.

These operation corresponding to the four possible combination of two classical bit

Classical bit	operation $\hat{\sigma}_A$	Result state
00	I	$ \phi^+\rangle$
01	$\hat{\sigma}_x$	$ \psi^+\rangle$
10	$\hat{\sigma}_z$	$ \phi^-\rangle$
11	$-i\hat{\sigma}_y$	$ \psi^-\rangle$

Final Bob decoding by BSA then the dense coding protocol is completed.

2. Show that the Bell states form an orthonormal basis for the two-qubit Hilbert space $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$

Solution

Four Bell states

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Show that $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$

1. Normalization (unit length)

$$\langle \Phi^\pm | \Phi^\pm \rangle = 1$$

$$\langle \Psi^\pm | \Psi^\pm \rangle = 1$$

$$\langle \Phi^\pm | \Phi^\pm \rangle = \langle \Psi^\pm | \Psi^\pm \rangle = 1$$

$$\langle \Phi^\pm | \Phi^\mp \rangle = \langle \Psi^\pm | \Psi^\mp \rangle = 0$$

2. Orthogonality

$$\langle \Phi^\pm | \Psi^\pm \rangle = 0$$

Example

Normalization
(unit length)

$$\langle \Phi^+ | \Phi^+ \rangle = \frac{1}{\sqrt{2}} \langle 00 | + \langle 11 | \otimes \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$= \frac{1}{2} (\langle 00 | 00 \rangle + 0 + 0 + \langle 11 | 11 \rangle)$$

$$= \frac{2}{2} = 1$$

$$= 1$$

Orthogonality (mutually perpendicular)

$$\langle \Phi^+ | \Psi^- \rangle = \frac{1}{\sqrt{2}} (\langle 00 | + \langle 11 |) \otimes \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle)$$

$$= \frac{1}{2} (\langle 00 | 01 \rangle - \langle 00 | 10 \rangle + \langle 11 | 01 \rangle - \langle 11 | 10 \rangle) = \frac{1}{2} (0 - 0 + 0 - 0)$$

$$= 0$$

$$= 0$$

Problem A: Non-cloning theorem (Bonus question)

14)

a) Given

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

$$|0\rangle = \text{fixed blank}$$

Here denote the initial state of the ancillary qubit similar to a blank paper in real copy machine

Solution

we have

$$\begin{aligned} |\psi\rangle \otimes |\psi\rangle &= (a|0\rangle + b|1\rangle) \otimes (a|0\rangle + b|1\rangle) \\ &= a^2|00\rangle + ab|01\rangle + ba|10\rangle + b^2|11\rangle \end{aligned}$$

The quantum copy machine cannot produce

$$\begin{aligned} \hat{C}(|\psi\rangle \otimes |0\rangle) &= \hat{C}((a|0\rangle + b|1\rangle) \otimes |0\rangle) \\ &= \hat{C}(a|00\rangle + b|10\rangle) \\ &= a\hat{C}|00\rangle + b\hat{C}|10\rangle \\ &= \underline{\underline{a|00\rangle + b|11\rangle}} \end{aligned}$$

Therefore

$$|\psi\rangle \otimes |\psi\rangle \neq \hat{C}(|\psi\rangle \otimes |0\rangle)$$

we see that the ideal copy and the machine-copied state

only 2

$$\alpha\beta = \beta\alpha = 0 \text{ and } \left. \begin{array}{l} \alpha = \alpha^2, \beta = \beta^2 \end{array} \right\}$$

which is the case for either $\alpha=0, \beta=1$ or $\alpha=1, \beta=0$

NB:

we only copy the simple computational basis state but not any superposition of them i.e not any general quantum state.

2) Classical bit

classical bits can be copied

Example

0000 $\rightarrow$ 0000, 0000

Encode bit of qubit

$$\begin{array}{l} 0 \rightarrow |0\rangle \\ 1 \rightarrow |1\rangle \end{array} \quad \text{where } |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Orthogonal basis state

The no-cloning theorem forbids cloning arbitrary qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

BUT $|0\rangle$ and $|1\rangle$ can be cloned

A cloning unitary for classical bits CNOT

$$\begin{array}{l} U(|0\rangle|0\rangle) = |0\rangle|0\rangle \\ U(|1\rangle|0\rangle) = |1\rangle|1\rangle \end{array}$$

Applied to a superposition

$$U((a|0\rangle + b|1\rangle)|0\rangle) = a|00\rangle + b|11\rangle$$

$$|u\rangle \otimes |v\rangle = (a|0\rangle + b|1\rangle)(a|0\rangle + b|1\rangle)$$

$$= \underline{(a|0\rangle + b|1\rangle)^2}$$

Info shows classical bit can be copied
Arbitrary quantum state cannot be copied

Classical bit Discrete-state - copy
- quantum state - not copy